

Badira — Week Plan: 20–26 September

2026-09-19 · @Someone

Where we stand

Round 1 of StartSmart is won. The final round is in Riyadh in about four weeks, so this is week 1 of 4.

The prototype at badira-ai.netlify.app already does the hard part well: full Arabic and English, a working Risk × Reliability × Time result, an explanation screen, and a doctor summary. What it does not yet have is a real answer behind the risk number — the result card still says **Simulated**.

That single word is the biggest exposure in Riyadh. A judge will ask what is behind the prediction. This week is about earning the right to answer that question:

1. First, understand how doctors actually decide a woman is at risk. We cannot model a decision we have not studied.
2. Then, find the data that would let us learn that decision.

Everything else waits.

Two tracks, three tasks

StartSmart judges a startup, not a paper. The goal splits in two, and both have to be true at the same time.

Track	The question judges ask	What answers it
Technical	Is the AI real, and is it safe?	A clinical grounding we can defend, a named dataset, an honest account of what the model does and does not do
Business	Why you, why here, why now?	The Saudi gap, Vision 2030 maternal-health framing, the Materna BP comparison, delivery of the pitch itself

This week's three tasks feed both tracks:

1. **Clinical detection deck** — technical foundation, and the credibility story for the business pitch.

2. **Dataset scouting** — pure technical, and the precondition for removing the word *Simulated*.
3. **Judge channel study** — pure business, and cheap. This is the bonus, not the main course.

Tasks 1 and 2 run in parallel, one student each. Task 3 is shared and light — no more than two hours for the week, each.

Task 1 — How doctors detect preeclampsia

Owner: Jwdee. Deliverable: a slide deck for Dr. Naila, due Thursday 24 September.

The point is not a literature review. The point is to draw the real clinical decision path, so that we know exactly which step Badira sits beside — and which steps it must never claim to replace.

The five questions the deck must answer

1. **What is preeclampsia, clinically?** New-onset hypertension after 20 weeks plus proteinuria or severe features. One slide, plain language, with the Arabic terms beside the English.
2. **How is a woman flagged as high-risk before anything goes wrong?** This is the risk-factor checklist route — history only, no lab, no scan. The [SMFM screening checklist](#) is the cleanest published version: six high-risk factors (previous preeclampsia, multifetal gestation, chronic hypertension, type 1 or 2 diabetes, renal disease, autoimmune disease) and nine moderate-risk ones (nulliparity, obesity, family history in mother or sister, low socioeconomic status, age ≥ 35 , prior low-birthweight or SGA infant, prior adverse pregnancy outcome, interpregnancy interval > 10 years, and a race factor). Aspirin is recommended with any one high-risk factor, and considered with more than one moderate factor — started 12–28 weeks, ideally before 16.
3. **What does the gold-standard screening actually measure?** The FMF competing-risks model at 11–14 weeks: maternal factors combined with mean arterial pressure, uterine artery Doppler, PIGF and PAPP-A. Explain what each one is and, importantly, what equipment and lab it needs.
4. **What actually happens in a Saudi antenatal clinic at 11–14 weeks?** Which of the measurements in question 3 are routinely available, and which are not. Ask Dr. Rania directly — this is the single most valuable slide in the deck and it cannot be found in a paper.
5. **Where does a woman fall through the gap?** The honest answer to “why does Badira need to exist.” One slide, and it becomes a pitch slide later.

Reading list

Start with the SMFM checklist above, then the ACOG hypertension-in-pregnancy practice bulletin, the NICE antenatal care guideline section on hypertension, the ASPRE trial (aspirin at 150 mg in screen-positive women), and one Saudi epidemiology paper for local prevalence and risk profile — search PubMed for *preeclampsia Saudi Arabia systematic review*, there is a recent meta-analysis.

Deck rules

- 12–15 slides. Not more.
- Every clinical claim carries its source on the slide.
- One diagram of the decision path, from booking visit to delivery, with the points where a risk decision is made marked.
- Last slide: **where Badira attaches to this path**. One arrow, one sentence.

That last slide is the whole reason for the task.

Task 2 — Find the data

Owner: Dana. Deliverable: a comparison spreadsheet plus a one-page recommendation, due Thursday 24 September.

Be realistic about what "done" means. In one week we will not have data in hand. We will have a ranked shortlist, an honest assessment of each option, and at least one access request actually sent.

Starting points already found

Dataset	Origin	Size	Preeclampsia recorded	Access
RAHMA — Riyadh Mother and Baby Cohort	Saudi — 3 Riyadh hospitals, 2013–15	14,568 women	Yes, 1.2% prevalence	Restricted; apply to the hospital ethics committees
Dr. Rania's hospital records	Saudi — local	Unknown, ask	Yes	Needs IRB; ask her this week
nuMoM2b (NICHD)	US — nulliparous cohort	~10,000	Yes	Application through the NICHD data repository — verify the process

Dataset	Origin	Size	Preeclampsia recorded	Access
UCI Maternal Health Risk	Bangladesh	1,013 rows, 6 features	No — generic risk level only	Open, immediate

RAHMA is the most interesting line in that table. It is Saudi, it is large, it records hypertensive disorders, and it collected exactly the kind of variables our Woman Schema asks for. It is also two years of paperwork away. Both facts matter.

The UCI set is the opposite: available in thirty seconds, and nearly useless for us — six vital signs, no history, no gestational age, no preeclampsia outcome. Many published papers use it anyway. Knowing why it does not fit is itself a strong answer to give a judge.

Also check, and do not skip

- **Kaggle maternal-health datasets.** Several exist. Most are copies of the UCI set or synthetic. Check provenance before trusting any of them — if you cannot find where the data came from, write that down and move on.
- **MIMIC-IV.** Large US critical-care database with obstetric admissions. Not built for this, but free and credentialed access is realistic for a student.
- **Arab and regional cohorts.** Search for Egyptian, Jordanian, Emirati and Qatari maternal cohorts. Qatar's Birth Cohort is worth a look.
- **The FMF / ASPRE model.** The model is published; the data is not. Still read it — it tells us which variables the best screening in the world actually uses.

The mapping — this is the real deliverable

Build one spreadsheet. Woman Schema fields down the rows, candidate datasets across the columns, and in each cell one of: **present**, **partial**, **absent**, or **derivable**.

Rows: age · nationality · gravidity · parity · previous preeclampsia · previous preterm birth · previous GDM · chronic hypertension · diabetes · kidney disease · autoimmune disease · BMI · blood pressure · gestational age at measurement · singleton or multiple · family history · outcome label · outcome timing.

Then add two summary rows at the bottom: **coverage** (what fraction of the schema the dataset fills) and **case count** (how many actual preeclampsia cases it contains). Those two numbers decide everything.

The recommendation page

One page, three parts: which dataset we build on now, which we apply for in parallel, and what we honestly say in Riyadh about where our data comes from. Write the third part as the sentence you would actually say out loud.

Task 3 — Study the judge's channel

Owner: both, two hours each across the week. Deliverable: one page, not a summary.

[Arbaaa Marketing](#) has 356,000 subscribers and 758 videos, in Arabic, on marketing, e-commerce, branding and AI. If he is on the panel, his published views tell us how he will listen. That is worth two hours.

Watch these five, in this order

Video	Why
أسباب فشل المشاريع: مشروعك ما فشل، أنت فشلت في توصيله! (9:02)	His view on why good projects lose. This is a pitch video in disguise.
دراسة السوق: كيف تدرس مكان مشروعك وتعرف منافسيك بدقة؟ (8:30)	We already have a competitor table. Check it against his method and fix whatever is missing.
استهداف العملاء: منتجك رهيب بس تباع للشخص الغلط! (8:51)	Who exactly is Badira for — the pregnant woman, the obstetrician, or the Ministry of Health? We have not decided.
الذكاء الاصطناعي في التسويق: فخ "النسخ والاصق" (7:07)	He is publicly sceptical of shallow AI. Expect that scepticism pointed at us.
زيادة المبيعات: متى يقرر العميل يشتري منتجك فعلاً؟ (11:38)	How he thinks about the moment of decision — useful for the demo's shape.

If there is time, the neuromarketing episode of بودكاست نهج (37,45:56k views) is his longest and most-watched piece.

One pattern to notice

Three separate videos argue that a founder's personal brand and reputation matter more than the company's. That is a consistent position, not a passing remark. It suggests he will be watching *you*, not only the slides. Two Saudi women engineers building a maternal-health product is a story — do not hide behind the architecture diagram.

What to hand in

Not summaries. One page with three headings:

1. **Three things he says a pitch must do** — in his words, with the video each came from.
2. **Where Badira's current story fails his test** — be harsh here. This page is only useful if it is uncomfortable.
3. **Two concrete changes to the pitch** — specific slides or sentences, not "improve the narrative".

This is preparation, not flattery. We are not going to quote him at himself on stage.

The week, day by day

Day	Jwdee — clinical	Dana — data
Sun 20	Read the SMFM checklist and the ACOG bulletin. Build the deck skeleton — headings only.	Open the four starting datasets. Build the empty mapping sheet with all schema rows.
Mon 21	Slides for questions 1 and 2: definition, risk-factor screening.	Fill the mapping for UCI and nuMoM2b. Read the RAHMA cohort profile in full.
Tue 22	Slides for question 3: the FMF 11–14 week model and what it requires.	Write the RAHMA access enquiry. Send it. Search for Arab and Gulf cohorts.
Wed 23	Meet Dr. Rania together. Question 4 for Jwdee, hospital-records feasibility for Dana.	Same meeting. Then fill the mapping row for Dr. Rania's records.
Thu 24	Deck finished and sent to Dr. Naila by evening.	Mapping sheet and recommendation page sent by evening.
Fri 25	Rest.	Rest.
Sat 26	Review with Dr. Naila. Swap deliverables and read each other's.	Same.

The judge's videos fit in the gaps — one on Sunday, two midweek, two before Saturday. They do not get their own slot.

Wednesday's meeting with Dr. Rania is the fixed point of the week. Book it first, before anything else. If it cannot happen Wednesday, move the whole week around it rather than dropping it.

What "done" looks like on Saturday

Five things on the table. If any one is missing, we have lost a week of four.

- ☐ A 12–15 slide deck on clinical detection, ending with one slide showing where Badira attaches to the care path
- ☐ A filled mapping spreadsheet: every Woman Schema field × every candidate dataset, with coverage and case-count totals
- ☐ A one-page dataset recommendation, including the sentence we will say out loud in Riyadh about our data
- ☐ At least one dataset access request actually sent, with a date
- ☐ One page on the judge's channel, with two concrete changes to the pitch

The test I will apply on Saturday

I will ask each of you one question: **"A judge says the risk score is fake. What do you say?"**

If the answer is vague, the week did not work, regardless of how good the documents look. The whole point of these five deliverables is to make that answer short, specific and true.

Not this week

Saying no is half the plan. None of the following happens before Saturday:

- **No new UI work.** The prototype is good enough. There are three small fixes waiting, and they take an afternoon in week 2.
- **No model training.** We do not have data. Training on the UCI set to have "a model" would be exactly the copy-paste trap, and it would not survive one question.
- **No pitch deck.** We cannot write the pitch before we know what we are claiming. Week 3.
- **No preterm birth or GDM.** They are in the research plan, not in the competition. Preeclampsia only.
- **No logo, colours or branding.** It already looks better than most of what will be on that stage.

The four weeks, roughly

1. **This week** — clinical grounding and data.
2. **Week 2** — build the risk engine on what we found, plus the three UI fixes.
3. **Week 3** — pitch deck, demo script, and the answers to hard questions.

4. **Week 4** — rehearse. Out loud, timed, in front of people who will interrupt.

Week 4 is rehearsal only. Protect it.